

05. Breast Cancer

Introduction

Breast cancer is one of the most common cancers affecting women worldwide, although it can also occur in men. It develops when abnormal cells in the breast grow uncontrollably and form a tumor. Early detection through regular screening significantly improves treatment success.

What is Breast Cancer?

Breast cancer begins in the cells of the breast, usually in the milk ducts or lobules. If left untreated, it can spread to nearby lymph nodes and other parts of the body through the bloodstream or lymphatic system.

Types of Breast Cancer

1. Ductal Carcinoma in Situ (DCIS)

A non-invasive cancer confined to the milk ducts.

2. Invasive Ductal Carcinoma (IDC)

The most common type that begins in the ducts and spreads into nearby breast tissue.

3. Invasive Lobular Carcinoma (ILC)

Begins in the milk-producing lobules and may spread to surrounding tissues.

4. Triple-Negative Breast Cancer

A more aggressive type that lacks estrogen, progesterone, and HER2 receptors.

5. HER2-Positive Breast Cancer

Characterized by excess HER2 protein and often treated with targeted therapy.

Risk Factors

- Increasing age
- Family history of breast cancer
- BRCA1 or BRCA2 gene mutations
- Previous breast cancer

- Early menstruation
 - Late menopause
 - Obesity
 - Alcohol consumption
 - Physical inactivity
 - Hormone replacement therapy
-

Symptoms

- Lump in the breast or underarm
 - Breast swelling
 - Change in breast shape or size
 - Nipple discharge
 - Bloody nipple discharge
 - Nipple inversion
 - Skin dimpling
 - Redness or thickened skin
 - Persistent breast pain
-

Diagnosis

Doctors may recommend:

- Clinical breast examination
 - Mammography
 - Breast ultrasound
 - Breast MRI
 - Needle biopsy
 - Hormone receptor testing
 - HER2 testing
-

Cancer Staging

Stage 0

Cancer remains inside the milk ducts.

Stage I

Small localized tumor.

Stage II

Larger tumor or limited lymph node involvement.

Stage III

Cancer has spread to nearby lymph nodes or chest wall.

Stage IV

Cancer has spread to distant organs such as the liver, lungs, bones, or brain.

Treatment Options

Surgery

- Lumpectomy
- Mastectomy
- Sentinel lymph node biopsy
- Axillary lymph node dissection

Chemotherapy

May be given before surgery (neoadjuvant) or after surgery (adjuvant).

Radiation Therapy

Uses high-energy rays to destroy remaining cancer cells after surgery.

Hormone Therapy

Used for hormone receptor-positive cancers.

Examples:

- Tamoxifen
- Aromatase inhibitors

Targeted Therapy

Examples include medications directed at HER2-positive cancers.

Immunotherapy

Recommended for selected patients with advanced or triple-negative breast cancer.

Possible Side Effects

- Hair loss
 - Fatigue
 - Nausea
 - Skin irritation
 - Arm swelling (lymphedema)
 - Infection risk
 - Menopausal symptoms
-

Recovery and Follow-up

Patients should:

- Attend regular follow-up appointments.
 - Perform breast self-awareness.
 - Maintain a healthy weight.
 - Exercise regularly.
 - Eat a balanced diet.
 - Avoid tobacco and excessive alcohol.
 - Report any new symptoms promptly.
-

Prevention

Although breast cancer cannot always be prevented, the risk may be reduced by:

- Maintaining a healthy body weight
- Regular physical activity
- Limiting alcohol

- Breastfeeding when possible
 - Participating in recommended screening programs
-

Frequently Asked Questions

Can men develop breast cancer?

Yes. Although uncommon, men can also develop breast cancer.

Is every breast lump cancer?

No. Many breast lumps are benign, but every new lump should be evaluated by a healthcare professional.

Can breast cancer be cured?

Many people with early-stage breast cancer are successfully treated, especially when diagnosed early.

How often should women undergo screening?

Screening recommendations vary by age and risk factors. Individuals should discuss the appropriate schedule with their healthcare provider.

Summary

Breast cancer is a common but often treatable disease, especially when detected early. Regular screening, awareness of symptoms, timely diagnosis, and appropriate treatment significantly improve outcomes and quality of life.